

Javad Noorbakhsh

Bioinformatics & Machine Learning Scientist

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SUMMARY

Computational biologist working on AI methods for cancer spatial omics and computational pathology, with a research background spanning cancer genomics and systems biology. Leads technical teams and cross-functional research collaborations.

EXPERIENCE

Bioinformatics & Machine Learning Scientist

Jul 2024 – Present

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Applied pathology foundation models to accurately identify and classify blood cell types within Giemsa-stained tissue slides.
- Engineered ML algorithms to cross-evaluate and compare spatial omics profiles between human and mouse models.
- Benchmarked the impact of H&E staining variations to optimize foundation model diagnostic performance.
- Evaluated spatial transcriptomics data from cancer patient cohorts to quantify therapeutic effects on microenvironment composition.
- Formulated mathematical models of tumor evolution and subclonal dynamics, enhancing precision in treatment strategy selection.
- Co-authored peer-reviewed reviews and competitive grant proposals advancing AI applications in spatial omics.

Principal Scientist

Apr 2022 – Feb 2024

Kojin Therapeutics, Boston, MA

- Spearheaded AI-driven multi-omics analysis, directly enabling fundraising and a pharmaceutical partnership.
- Established high-throughput assay screening pipelines, identifying novel small-molecule hits for drug development.
- Deployed graph neural networks (GNNs) for molecular design, accelerating target discovery and experimental drug candidate validation.
- Architected attention-based multi-instance learning pipelines on histology images, uncovering potential drug-sensitive cancer subtypes.

Lead Scientist

Jan 2020 – Apr 2022

Broad Institute of MIT and Harvard, Cambridge, MA

- Orchestrated the Cancer Dependency Map omics team, managing end-to-end RNAseq, WES, and WGS pipelines to deliver quarterly releases across public, institutional, and commercial portals.
- Standardized and automated production pipelines across major projects, achieving reproducible, timely data distribution.
- Directed ML initiatives aligning patient tumor NGS data with cell lines, boosting alignment accuracy and tool reliability.

Research Scientist

Oct 2017 – Dec 2019

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Pioneered a cloud-based deep learning framework for tumor state classification across histopathology images, attaining AUCs > 0.9 across 19 cancer types.

- Guided the development and deployment of an image processing framework for 3D organoid segmentation, producing an open-source software package.

Postdoctoral Fellow

Feb 2015 – Oct 2017

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Investigated copy number variations and mutation calling dynamics in cancer, identifying key biomarker correlations with patient survival.
- Constructed mathematical models of clonal evolution, improving quantitative assessment of genetic heterogeneity across diverse tumor types.

EDUCATION

Ph.D., Physics — Boston University, Boston, MA

Sep 2009 – Dec 2014

M.Sc., Physics — Sharif University of Technology, Tehran, Iran

Sep 2006 – Dec 2008

B.Sc., Electrical Engineering — Amirkabir University of Technology, Tehran, Iran

Sep 2001 – Dec 2005

SKILLS

Quantitative Analysis Machine Learning, Deep Learning, Bioinformatics, Statistics, Image Processing, Mathematical Modeling, Cheminformatics

Programming Python, TensorFlow/PyTorch, Scikit-learn, R, Bash, HPC, Cloud Computing, Containerization, Kubernetes, Version Control, C++, MATLAB

Leadership People Management, Mentorship, Team Collaboration

PUBLICATIONS

Reverse chronological. Author lists abbreviated with “et al.”; full record on [Google Scholar](#).

1. **Prediction of outcome from spatial Protein profiling of triple-negative breast cancers.** Ali Foroughi Pour, Te-Chia Wu, **Javad Noorbakhsh**, Jan Martinek, Santhosh Sivajothi, et al. *Communications Medicine*, 2026.
2. **RANKL inhibition spatially rewires the microenvironment of luminal breast cancer.** Mario Rodriguez-del Collado, Andrea Vethencourt, Alexandra Barranco, Jaime Martinez de Villarreal, **Javad Noorbakhsh**, et al. *medRxiv*, 2026.
3. **DIANNE: Segmentation-Free Localization of Histology Differential Attributes.** Sergii Domanskyi, Jill C Rubinstein, Todd B Sheridan, Adam Thiesen, **Javad Noorbakhsh**, et al. *bioRxiv*, 2026.
4. **Emerging AI approaches for cancer spatial omics.** **Javad Noorbakhsh**, Ali Foroughi Pour, Jeffrey Chuang. *GigaScience*, 2025.
5. **Germline variation contributes to false negatives in CRISPR-based experiments with varying burden across ancestries.** Sean A Misek, Aaron Fultineer, Jeremie Kalfon, **Javad Noorbakhsh**, Isabella Boyle, et al. *Nature Communications*, 2024.
6. **The enzyme glutamate-cysteine ligase (GCL) is a target for ferroptosis induction in cancer.** John K Eaton, Priya Chatterji, Laura Furst, Sneha Basak, **Javad Noorbakhsh**, et al. *bioRxiv*, 2024.
7. **High-throughput deconvolution of 3D organoid dynamics at cellular resolution for cancer pharmacology with Cellos.** Patience Mukashyaka, Pooja Kumar, David J Mellert, Shadae Nicholas, **Javad Noorbakhsh**, et al. *Nature Communications*, 2023.
8. **Computational estimation of quality and clinical relevance of cancer cell lines.** Lucia Trastulla, **Javad Noorbakhsh**, Francisca Vazquez, James McFarland, Francesco Iorio. *Molecular Systems Biology*, 2022.
9. **Bridging the gap between cancer cell line models and tumours using gene expression data.** **Javad Noorbakhsh**, Francisca Vazquez, James M McFarland. *British Journal of Cancer*, 2021.

10. **Deep learning-based cross-classifications reveal conserved spatial behaviors within tumor histological images.** Javad Noorbakhsh, Saman Farahmand, Sandeep Namburi, Dennis Caruana, David Rimm, et al. *Nature Communications*, 2020.
11. **Repeat expansions confer WRN dependence in microsatellite-unstable cancers.** Niek van Wietmarschen, Sriram Sridharan, William J Nathan, Anthony Tubbs, Javad Noorbakhsh, et al. *Nature*, 2020.
12. **Treating cancer as an invasive species.** Javad Noorbakhsh, Zi-Ming Zhao, James C Russell, Jeffrey H Chuang. *Molecular Cancer Research*, 2020.
13. **Integrative deep learning for PanCancer molecular subtype classification using histopathological images and RNAseq data.** Fatima Zare, Javad Noorbakhsh, Tianyu Wang, Jeffrey H Chuang, Sheida Nabavi. *ACM-BCB 2020*, 2020.
14. **Longitudinal molecular trajectories of diffuse glioma in adults.** Floris P Barthel, Kevin C Johnson, Frederick S Varn, Anzhela D Moskalik, Javad Noorbakhsh, et al. *Nature*, 2019.
15. **CCNE1 amplification is associated with poor prognosis in patients with triple negative breast cancer.** Zi-Ming Zhao, Susan E Yost, Katherine E Hutchinson, Sierra Min Li, Javad Noorbakhsh, et al. *BMC Cancer*, 2019.
16. **Machine learning in biology and medicine.** Javad Noorbakhsh, Harshpreet Chandok, R Krishna Murthy Karuturi, Joshy George. *Advances in Molecular Pathology*, 2019.
17. **Distribution-based measures of tumor heterogeneity are sensitive to mutation calling and lack strong clinical predictive power.** Javad Noorbakhsh, Hyunsoo Kim, Sandeep Namburi, Jeffrey H Chuang. *Scientific Reports*, 2018.
18. **High-resolution deconstruction of evolution induced by chemotherapy treatments in breast cancer xenografts.** Hyunsoo Kim, Pooja Kumar, Francesca Menghi, Javad Noorbakhsh, Eliza Cerveira, et al. *Scientific Reports*, 2018.
19. **Uncertainties in tumor allele frequencies limit power to infer evolutionary pressures.** Javad Noorbakhsh, Jeffrey H Chuang. *Nature Genetics*, 2017.
20. **From intracellular signaling to population oscillations: bridging size-and time-scales in collective behavior.** Allyson E Sgro, David J Schwab, Javad Noorbakhsh, Troy Mestler, Pankaj Mehta, et al. *Molecular Systems Biology*, 2015.
21. **Modeling oscillations and spiral waves in Dictyostelium populations.** Javad Noorbakhsh, David J Schwab, Allyson E Sgro, Thomas Gregor, Pankaj Mehta. *Physical Review E*, 2015.
22. **Two-dimensionality of yeast colony expansion accompanied by pattern formation.** Lin Chen, Javad Noorbakhsh, Rhys M Adams, Joseph Samaniego-Evans, Germaine Agollah, et al. *PLoS Computational Biology*, 2014.
23. **Intrinsic noise of microRNA-regulated genes and the ceRNA hypothesis.** Javad Noorbakhsh, Alex H Lang, Pankaj Mehta. *PLoS One*, 2013.

PATENT

Convolutional neural networks for classification of cancer histological images. Jeffrey Hsu-Min Chuang, Javad Noorbakhsh, Kourosh Zarrinhalam, Saman Farahmand, Mohammad Soltanieh-ha. *U.S. Patent Application US 2025/0378559 A1*, 2025.